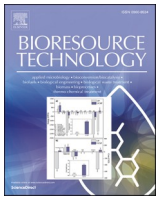
Bioresource Technology 402 (2024) 130787



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Bioresource Technology

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Enhanced anaerobic digestion performance and sludge filterability by membrane microaeration for anaerobic membrane bioreactor application

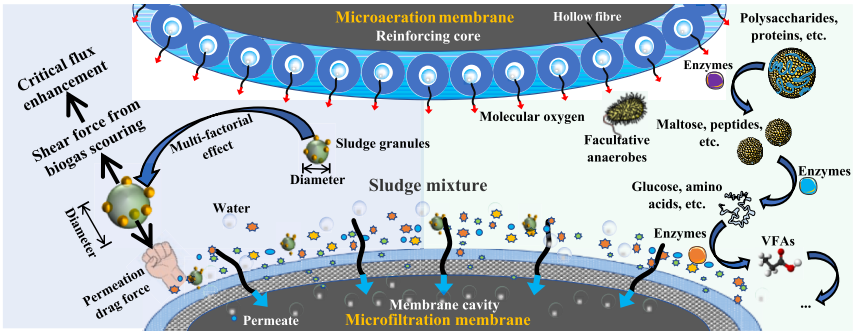
Chun-Hai Wei [a,b,1,*], Zheng-Wei Wang [a,1], Jian-Hua Dai [a], Kang Xiao [c], Hua-Rong Yu [a,b], Fang-Shu Qu [a,b], Hong-Wei Rong [a,b], Jun-Guo He [a,b], Huu Hao Ngo [d]

[a] Department of Municipal Engineering, School of Civil Engineering, Guangzhou University, Guangzhou 510006, China
[b] Key Laboratory for Water Quality and Conservation of the Pearl River Delta, Ministry of Education, Guangzhou 510006, China
[c] Beijing Yanshan Earth Critical Zone National Research Station, College of Resources and Environment, University of Chinese Academy of Sciences, Beijing 101408, China
[d] Centre for Technology in Water and Wastewater, University of Technology, Sydney NSW 2007, Australia

HIGHLIGHTS

- Bubble-free membrane microaeration was imported into anaerobic membrane bioreactor.
- Microaeration in AD led to 65.7–144.8% more methane yield from protein and starch.
- Microaeration raised critical flux of MF membrane by driving large flocs formation.
- Microaeration enriched functional genes and bacteria for dissolution/hydrolysis.
- Microaeration showed little negative effect on functional genes for methanogenesis.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Membrane fouling potential
Methanogenesis
Oxidation-reduction potential
Particle size distribution
Volatile fatty acid

ABSTRACT

Slow dissolution/hydrolysis of insoluble/macromolecular organics and poor sludge filterability restrict the application potential of anaerobic membrane bioreactor (AnMBR). Bubble-free membrane microaeration was firstly proposed to overcome these obstacles in this study. The batch anaerobic digestion tests feeding insoluble starch and soluble peptone with and without microaeration showed that microaeration led to a 65.7–144.8% increase in methane production and increased critical flux of microfiltration membrane via driving the formation of large sludge flocs and the resultant improvement of sludge settleability. The metagenomic and bioinformatic analyses showed that microaeration significantly enriched the functional genes and bacteria for polysaccharide and protein hydrolysis, microaeration showed little negative effects on the functional genes involved in anaerobic metabolisms, and substrate transfer from starch to peptone significantly affected the functional genes and microbial community. This study demonstrates the dual synergism of microaeration to enhance the dissolution/hydrolysis/acidification of insoluble/macromolecular organics and sludge filterability for AnMBR application.

* Corresponding author at: Department of Municipal Engineering, School of Civil Engineering, Guangzhou University, Guangzhou 510006, China.
E-mail address: weich@gzhu.edu.cn (C.-H. Wei).
[1] Equal contribution

https://doi.org/10.1016/j.biortech.2024.130787
Received 1 February 2024; Received in revised form 30 April 2024; Accepted 1 May 2024
Available online 3 May 2024
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1. Introduction

Anaerobic digestion (AD) is promising to treat high-strength organic wastewater and solid waste for organics removal and methane recovery (Zhang et al., 2020; Zhang et al., 2021a). However, high-concentration insoluble organics in some wastewaters (e.g., livestock wastewater) and solid wastes (e.g., excess sludge) usually take long time (20–30 d) for sequential dissolution/hydrolysis, acidogenesis, acetogenesis, and methanogenesis in conventional completely mixed AD process. Anaerobic membrane bioreactor (AnMBR) coupling anaerobic activated sludge degradation and micro/ultra-filtration membrane separation can achieve complete separation of hydraulic and sludge retention time and partial separation of hydraulic and macromolecular organics retention time via high-efficiency membrane rejection, being an advanced process suitable for treating wastewater containing high-concentration insoluble organics (Ying et al., 2021; Yu et al., 2016). However, AnMBR is still facing the challenges including slow dissolution/hydrolysis rate of insoluble organics, sensitive methanogenic activity and membrane fouling caused by poor anaerobic sludge filterability (Song et al., 2018).

Microaeration (i.e., injecting or in-situ chemically/electrochemically generating trace oxygen) has been considered as an effective strategy to enhance hydrolysis in AD and remove hydrogen sulfide from biogas in recent years (Hu et al., 2017). The trace oxygen typically at 0.005–1 L O_2/L reactor/day into AD system would be rapidly consumed by facultative bacteria (Nguyen & Khanal, 2018), which could not only stimulate the secretion of extracellular enzymes and thus promote the dissolution/hydrolysis of insoluble/macromolecular organics, but also protect methanogens from oxygen stress and thus avoid negative effects on methanogenesis (Fu et al., 2022). Several studies (Amin et al., 2021; Cao et al., 2022; Iftikhar et al., 2021) evaluated microaeration-assisted AD for swine/cattle manure and cellulose treatment and showed that microaeration effectively improved the dissolution/hydrolysis efficiency of insoluble organics and finally enhanced methane production. Microaeration may facilitate some microorganisms with a unique *cytochrome bd* oxidase to achieve nanaerobic respiration at very low dissolved oxygen (<2 μM), where simple organics (e.g., glucose, butyrate, propionate, acetate) can be directly oxidized to carbon dioxide (Wu et al., 2020). Hydrogenotrophic methanogens can further convert these carbon dioxide and hydrogen derived from acidogenesis and acetogenesis into methane (Wu et al., 2023). Microaeration has also some advantages of simple operation and low cost over other reported oxygen supply methods for AD including in-situ slow oxygen release via dosing calcium peroxide (Fu et al., 2018) and in-situ oxygen supply via water electrolysis (Tartakovsky et al., 2011), thus promoting its real application.

Microaeration in AD may induce the formation of stratified flocs with a core of strict anaerobes (e.g., methanogens) and an outer layer of facultative bacteria, where the outer layer of facultative bacteria protects the oxygen-sensitive anaerobes by scavenging reactive oxygen species through aerobic metabolism (Liu et al., 2009; Nguyen & Khanal, 2018). This may also affect the secretion of extracellular polymeric substances (EPS) and induce the formation of larger flocs, which further affect the settleability and dewaterability of anaerobic sludge (Stewart & Franklin, 2008). This was confirmed in a recent study (Zhou et al., 2021), which found that in situ microaeration (aeration of 12.5 mL/L/d) not only increased methane production and volatile solids removal, but also effectively increased sludge particle size and improved its dewaterability via reducing the ratio of loosely-bounded EPS to tightly-bounded EPS. In addition, the formation of large microbial aggregates with outer facultative bacteria and inner methanogens may not only facilitate the potential syntrophic interactions between bacteria and methanogens, but also improve its settleability and thus mitigate the potential membrane fouling in AnMBR application (Cheng et al., 2018).

Although microaeration has shown the abovementioned advantages for AD treatment of high-concentration insoluble organic wastewater and solid waste, some drawbacks still hinder its wide application (Aslam et al., 2021; Fu et al., 2022), for example, how to achieve uniform aeration, improve the oxygen transfer rate and achieve the high-efficiency synergy between microaeration and methanogenesis is a key consideration (Chen et al., 2019). Bubble-free membrane microaeration can directly supply molecular oxygen into anaerobic sludge mixture and thus eliminates the gas-to-liquid mass transfer resistance, which has been used to achieve more uniform aeration and enhance refractory compound's hydrolysis (Zhang et al. 2022a). Therefore, a new AnMBR prototype assisted with membrane microaeration was proposed for the first time in this study. Sequential batch AD tests without, with, and without oxidation–reduction potential (ORP) controlled microaeration were conducted using two substrates (insoluble starch and soluble peptone), where organics degradation, methane production and microbial metagenomic characteristics were monitored in order to clarify the performance and mechanism of microaeration on AD of insoluble and macromolecular organics (see supplementary materials). At the end of each batch AD test, both critical flux evaluation of the microfiltration membrane and sludge characterization (suspended solids fractionation and particle size distribution) were conducted to reveal the performance and mechanism of microaeration on sludge filterability. The outcomes of this study would provide new insights into the potential double enhancements for AnMBR application by microaeration.

2. Materials and methods

2.1. Reactor and raw anaerobic sludge

The AnMBR prototype used in this study (see supplementary materials) consisted of a completely mixing anaerobic bioreactor with an effective volume of 22.5 L, a submerged polyvinylidene fluoride (PVDF) hollow fiber microfiltration membrane module with an effective filtration area of 0.34 m^2 and nominal pore size of 0.1 μm (Beijing Origin Water, China), a ZeeLung selective oxygen permeable bubble-free aeration hollow fiber membrane module with an effective area of 0.25 m^2 (Veolia, France, see supplementary materials), a peristaltic suction pump for membrane permeating, a biogas circulation pump supplying biogas at a recirculation rate of 15 L/min via a perforated tube to the bottom of two membrane modules and thus creating well-mixing in the reactor and scouring on membrane surface, a microaeration pump with pressure adjustment (0–5 kPa) depending on ORP to deliver air into the lumen of hollow fiber aeration membrane, pressure gauges for monitoring suction pressure of microfiltration membrane and air supply pressure of microaeration membrane, and a AMPTS II fully automatic methane potential testing system for recording methane production (Bioprocess Control, Sweden, see supplementary materials). The anaerobic activated sludge in the reactor had been well acclimated to and grew with the soluble starch (i.e., dissolving the insoluble starch mentioned below in hot water of 70 °C) as carbon source during a 6-month AnMBR operation prior to this study and its characteristics at the beginning of this study were described (see supplementary materials).

2.2. Batch AD tests

Sequential batch AD tests without microaeration (AD1) → with microaeration (AD2) → with microaeration (AD3) and without microaeration (AD4) → with microaeration (AD5) → with microaeration (AD6) were conducted using two representative carbon sources – insoluble starch and soluble bacterial peptone prepared by enzymatic digestion of selected animal tissues (i.e., a mixture of proteins, peptides and amino acids), respectively (see supplementary materials). At the start of each AD test, the concentrated stock of insoluble starch or soluble peptone was directly injected into the bioreactor to reach the target dosage of 4 g/L for chemical oxygen demand (COD), NH_4Cl and Na_2HPO_4 in the concentrate stock were used as nitrogen and phosphorus sources to ensure the target COD:N:P of 200:5:1 for the added substrates,

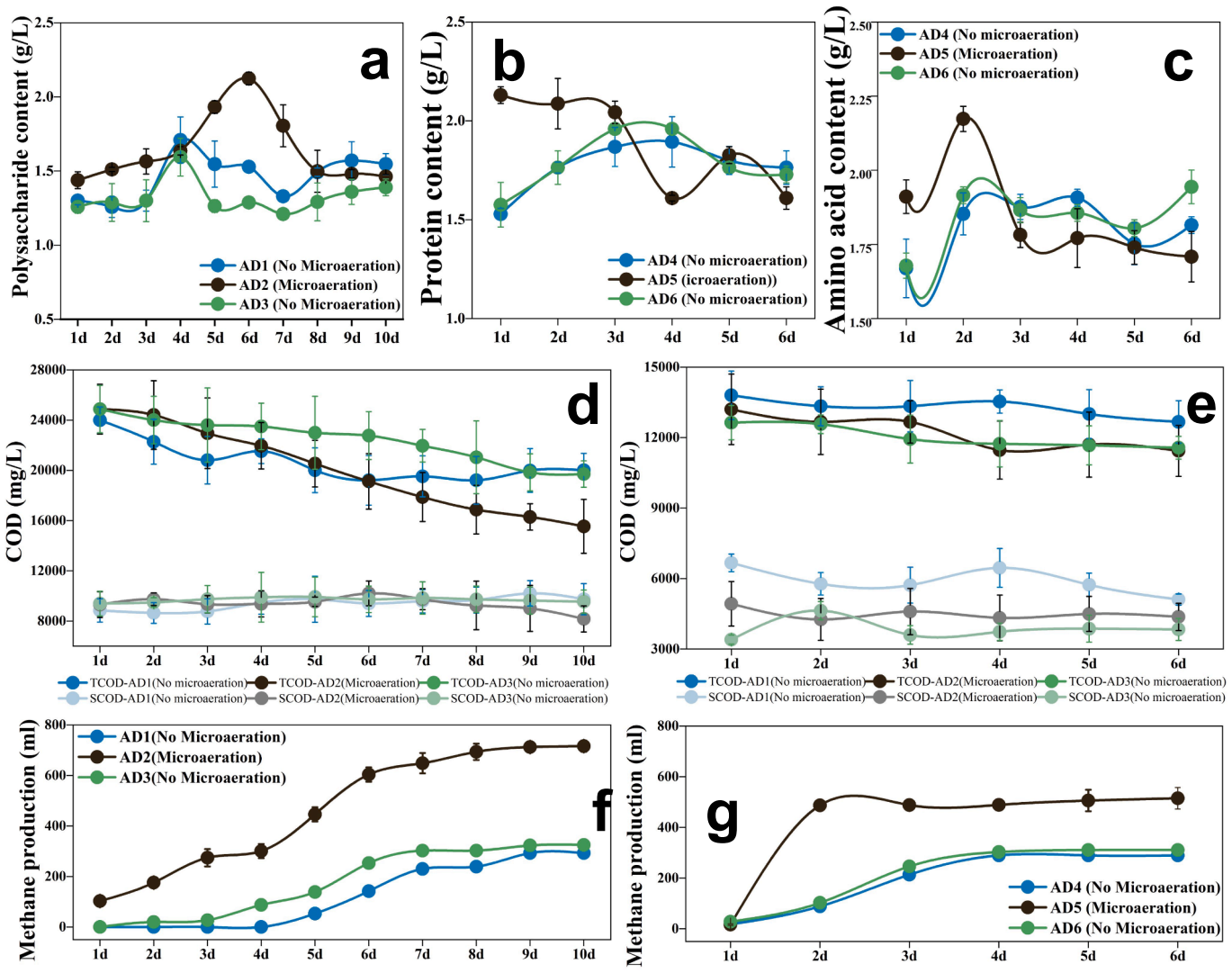


Fig. 1. Polysaccharide in starch-feeding group (a), protein (b) and amino acid (c) in peptone-feeding group, TCOD/SCOD in starch-feeding group (d) and peptone-feeding group (e), methane production in starch-feeding group (f) and peptone-feeding group (g).

$NaHCO_3$ in the concentrate stock was also dosed at 4 g/L (i.e., equivalent to COD) as a buffer to avoid the large fluctuation of pH in the bioreactor. At the end of each AD test, the supernatant was discharged via microfiltration membrane permeation to make the effective volume as the same with before injecting the concentrated stock. In order to avoid any oxygen intrusion for AD tests without microaeration, nitrogen stripping via the perforated tube was conducted to flush the system immediately after injecting the concentrated stock. The ORP was controlled in the range of −150 to −100 mV via adjusting the output of microaeration pump in AD2 and AD5, while it was in the range of −400 to −300 mV in other AD tests without microaeration. The pH was in the range of 6.7–8.1 during all AD tests without extra acid/alkali dosage for pH control. All AD tests were conducted in room temperature of 20–22 °C. During each batch test, sludge sample of 10 ml was taken daily and prefiltrated via 0.45 μm membrane syringe filter for water quality determination. In addition, the half of anaerobic sludge in the bioreactor was taken out for seeding another reactor after AD3 and the same amount of tap water was filled to the bioreactor before AD4, thus resulting in the much lower initial COD in AD4-6 than AD1-3. Methane production without microaeration was directly measured by AMPTS II, while methane production with microaeration was collected via gas bag and measured by gas chromatography (Techcomp, GC7900, China) to eliminate the effect of entrapped nitrogen via membrane microaeration (Amin et al., 2021). Each batch test was operated till no methane production.

2.3. Sludge filterability tests via critical flux determination

At the end of each AD test, the critical flux of microfiltration membrane was measured via the classical flux stepwise increment methodology (Fox & Stuckey, 2015; Le Clech et al. 2003b), where transmembrane pressure (TMP) as membrane fouling indicator was continuously measured with time under each constant flux and TMP increasing rate was employed to determine critical flux zone. Membrane permeate was returned into the bioreactor and the same peristaltic pump speed was employed during filtration test to make the results comparable among different AD tests in this study. During critical flux determination, a 15 min idle interval for the microfiltration membrane without permeation was employed between two constant-flux filtration tests in order to remove the deposited particles on membrane surface via biogas scouring. During AD tests, the microfiltration membrane without permeation was always under biogas scouring condition. Sludge was also sampled for solids characterization in terms of easy/difficult settleability and particle size distribution to clarify the potential mechanism behind the critical flux changes without and with microaeration.

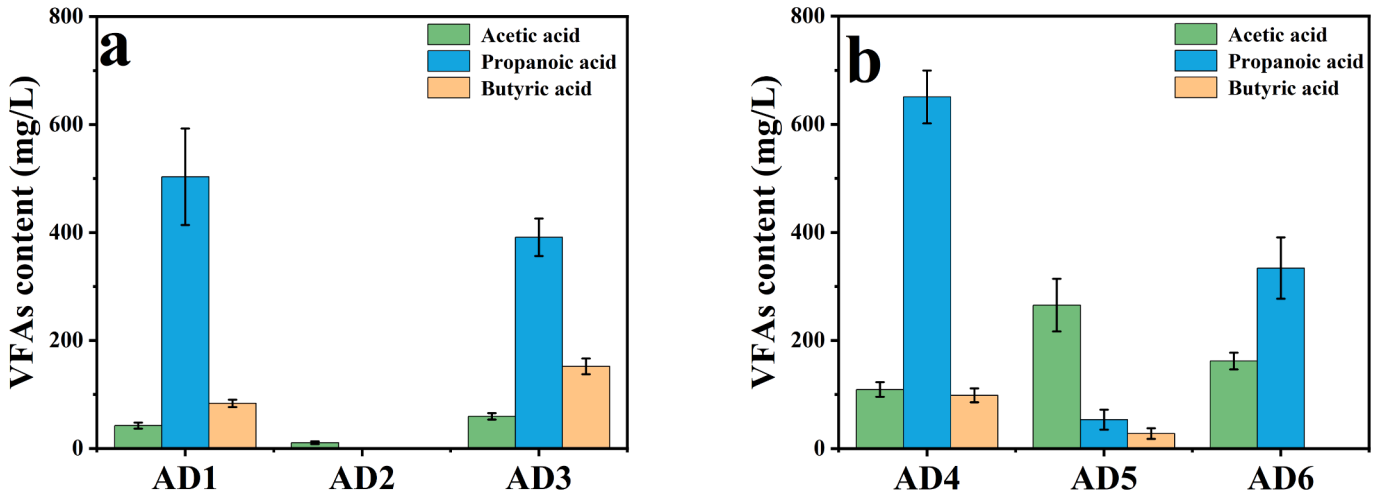


Fig. 2. VFAs concentration at the end of AD test in starch-feeding group (a) and peptone-feeding group (b).

2.4. Physicochemical analyses

Sludge samples were directly measured for total solids (TS), suspended solids (SS), volatile suspended solids (VSS), and total chemical oxygen demand (TCOD) according to the standard methods (Federation & Association, 2005). Sludge samples were separated by centrifugation at 3000 g for 10 min into pellet (i.e., easy-to-settle SS) and supernatant (i.e., difficult-to-settle SS). The laser particle size analyzer (BT-9300S, Dandong Baite Instrument Co., Ltd, China) was used to measure particle size distribution of sludge mixtures (including original sludge, pellet, and supernatant) based on dynamic light scattering method. The supernatant samples were further separated by 0.45 μm membrane syringe filter to produce water samples. The water samples were measured for soluble chemical oxygen demand (SCOD) via the standard methods (Federation & Association, 2005), for proteins via the Bradford method, for amino acids via the ninhydrin colorimetric method, for polysaccharides via the phenol–sulphuric acid method, for volatile fatty acids (VFAs) via gas chromatography (Techcomp, GC7980, China), and for fluorescent substances via the three-dimensional fluorescence spectrometry (Shimadzu-RF-6000, Japan), respectively.

2.5. Metagenomics analyses

Metagenome sequencing and bioinformatics analysis were employed to investigate microbial communities and critical function genes related to AD via the Majorbio Platform (Zhang et al., 2019). The sludge mixture was firstly concentrated by centrifugation (4000 g, 5 min) to obtain a microbial-rich pellet sample for testing, which was stored at −80 °C pending delivery. Three samples from raw sludge before AD tests (notated as Sludge-Blank), starch-feeding sludge at the end of AD2 with microaeration (notated as Sludge-Starch), and peptone-feeding sludge at the end of AD5 with microaeration (noted as Sludge-Peptone) were obtained for the metagenome sequencing analysis. The brief process included: 1) Total genomic DNA was extracted from the samples using the E.Z.N.A.® Soil DNA Kit (Omega Bio-tek, Norcross, GA, U.S.); 2) DNA extract quality was checked on 1 % agarose gel; 3) Fragmentation of samples (~400 bp) was conducted using ultrasound (Covaris M220, USA); 4) Sequencing library was constructed using the NEXTflex Rapid DNA-Seq; 5) Sequencing was finally conducted via the Illumina NovaSeq/Hiseq Xten (Illumina Inc., San Diego, CA, USA) by Majorbio BioPharm Technology Co., Ltd. (Shanghai, China). Raw data for genome sequencing was processed on a free online platform at Majorbio Cloud (**cloud.majorbio.com**). Species metabolic pathway-related genes were obtained from the Kyoto Encyclopedia of Genes and Genomes (KEGG) database (**https://www.genome.jp/kegg/**).

3. Results and discussion

3.1. Effect of microaeration on AD performance

3.1.1. Organics removal and methane recovery

Fig. 1a, d, and f shows the soluble polysaccharide, TCOD/SCOD, and methane production with reaction time for the starch-feeding groups (i.e., AD 1–3), respectively. From the Fig. 1a, soluble polysaccharides showed an initial increase followed by decrease with reaction time, indicating the dissolution of the injected insoluble starch and subsequent utilization by hydrolytic bacteria. There was a 67.7–104 % higher initial increase and 244.5–307 % higher subsequent decrease of soluble polysaccharides in AD2 with microaeration than in AD1 and AD3 without microaeration, showing the significantly enhanced dissolution/hydrolysis by microaeration, which was consistent with previous studies (Hu et al., 2017; Zhang et al. 2022b). From Fig. 1d, TCOD showed a gradual decrease while SCOD showed small fluctuations with reaction time, indicating that the insoluble starch was the main contributor of TCOD removal and the dissolution of insoluble starch was the key rate-limiting step for its anaerobic degradation. There was a 80.7–125.8 % higher TCOD removal in AD2 with microaeration than in AD1 and AD3 without microaeration, showing the significantly enhanced insoluble starch removal by microaeration. From Fig. 1f, methane production showed a lag period of 4 and 3 d for AD1 and AD3 without microaeration, respectively, which could be derived from slow dissolution of insoluble starch. Then, methane production showed a rapid increase till 7 d followed by a slow increase to 292.6 and 324.9 ml at the end of AD1 and AD3, respectively. In AD2 with microaeration, methane production showed a rapid increase till 6 d followed by a slow increase to a final 716 ml. No lag period for methane production and a 144.7 % and 120.4 % higher methane production in AD2 with microaeration than AD1 and AD3 without microaeration demonstrated that microaeration did not inhibit methanogenesis but on the contrary promoted methanogenesis via enhancing the dissolution/hydrolysis of insoluble starch in this study, which was consistent with previous studies (Hu et al., 2017; Zhang et al. 2022b).

Fig. 1b, c, e, and g shows protein, amino acid, COD, and methane production with reaction time for the peptone-feeding groups (i.e., AD 4–6), respectively. From Fig. 1b, there was an initial protein increase followed by a slow protein decrease with reaction time in AD4 and AD6 without microaeration, indicating the possible initial microbial secretion of endogenous protein in terms of soluble microbial products and the following protein hydrolysis with reaction time. There was a 35.2–39.4 % higher initial protein in AD5 with microaeration than AD4 and AD6 without microaeration, implying that microaeration

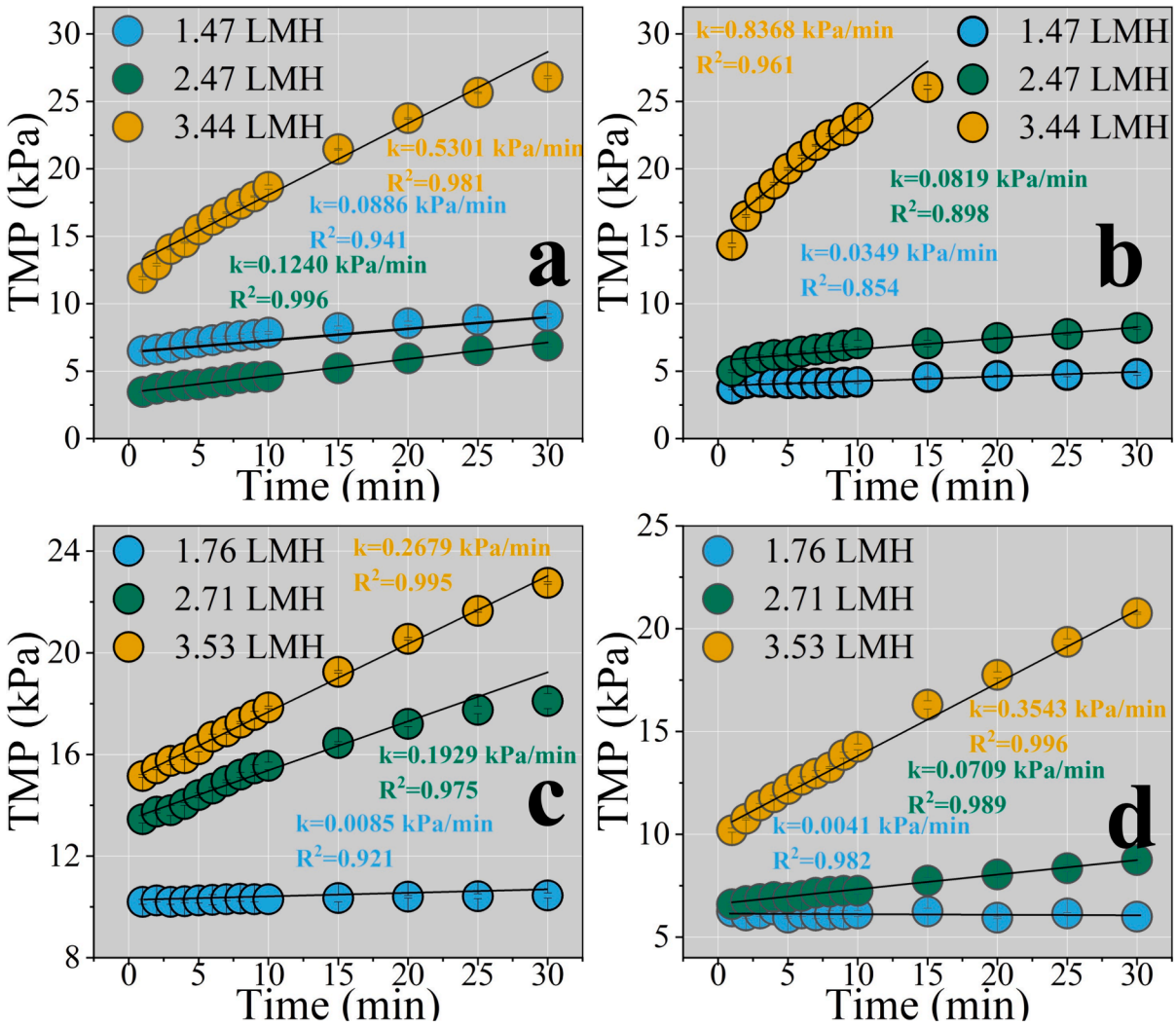


Fig. 3. Critical flux test at the end of AD1 (a) and AD2 (b) for starch-feeding group and of AD4 (c) and AD5 (d) for peptone-feeding group.

significantly enhanced the secretion of endogenous protein. In addition, there was a 108 %-255.6 % higher protein removal in AD5 with microaeration than AD4 and AD6 without microaeration, indicating the significantly enhanced protein removal by microaeration (Zhang et al., 2021b). From Fig. 1c, there was a 9.9 % higher initial increase and 318.4–401.7 % higher subsequent decrease of amino acid with reaction time in AD5 than in AD4 and AD6, showing the enhanced protein hydrolysis and amino acid utilization by microaeration. From Fig. 1e, there was a slow decrease of TCOD and SCOD with reaction time, and TCOD removal increased by 58.5 %-63 % in AD5 with microaeration than in AD4 and AD6 without microaeration, indicating the significantly enhanced protein mineralization by microaeration. From Fig. 1g, methane production without lag period showed an initial rapid increase followed by slow increase to the final 289.7 ml, 515.1 ml, 310.9 ml in AD4, AD5, AD6, respectively. Significantly faster and a 77.8 % and 65.7 % higher methane production in AD5 with microaeration than in AD4 and AD6 without microaeration demonstrated that microaeration did not inhibit methanogenesis but on the contrary promoted methanogenesis via enhancing protein hydrolysis and subsequent amino acid utilization in this study. The rapid methane production from the 1st to 2nd day could be derived from the consumption of peptides, while the changes in protein and amino acid with reaction time might be affected by the secretion of endogenous soluble microbial products from microbial metabolism and their decrease after the rapid methane production might be ascribed into microbial cell anabolism rather than methane production catabolism. In addition, from fluorescence analysis (see supplementary materials), the intensity of the protein peaks decreased from the start to the end of AD test significantly in AD5, but not in AD4, and decreased slightly in AD6, which were similar to the results for protein and amino acid, demonstrating the enhanced hydrolysis of protein-like substances by microaeration.

3.1.2. *Residual VFAs distribution*

Fig. 2a and b shows the VFAs concentration at the end of AD tests with starch and peptone feeding, respectively. From Fig. 2a for starch-feeding groups, AD2 with microaeration contained only 10.6 mg/L of acetic acid, while AD1 and AD3 without microaeration contained propionic acid (503.3 and 391.3 mg/L) and butyric acid (83.6 and 151.9 mg/L), respectively. From Fig. 2b for peptone-feeding groups, AD5 with microaeration contained higher concentration of acetic acid (265.3 mg/L), while AD4 and AD6 without microaeration contained higher concentration of propionic acid (650.6 and 333.9 mg/L, respectively). Acetic acid seemed to dominate under microaeration conditions, and it could be consumed more rapidly by methanogens. Propionic acid appeared to be more abundant without microaeration conditions, which might be a result of the high abundance of *Propionibacterium* (a fast growing and producing propionic acid species under exclusively anaerobic conditions) (Sawatdeenarunat et al., 2017). Propionic acid metabolism was known to be thermodynamically unfavorable, while microaeration seemed to avoid propionic acid production via inhibiting

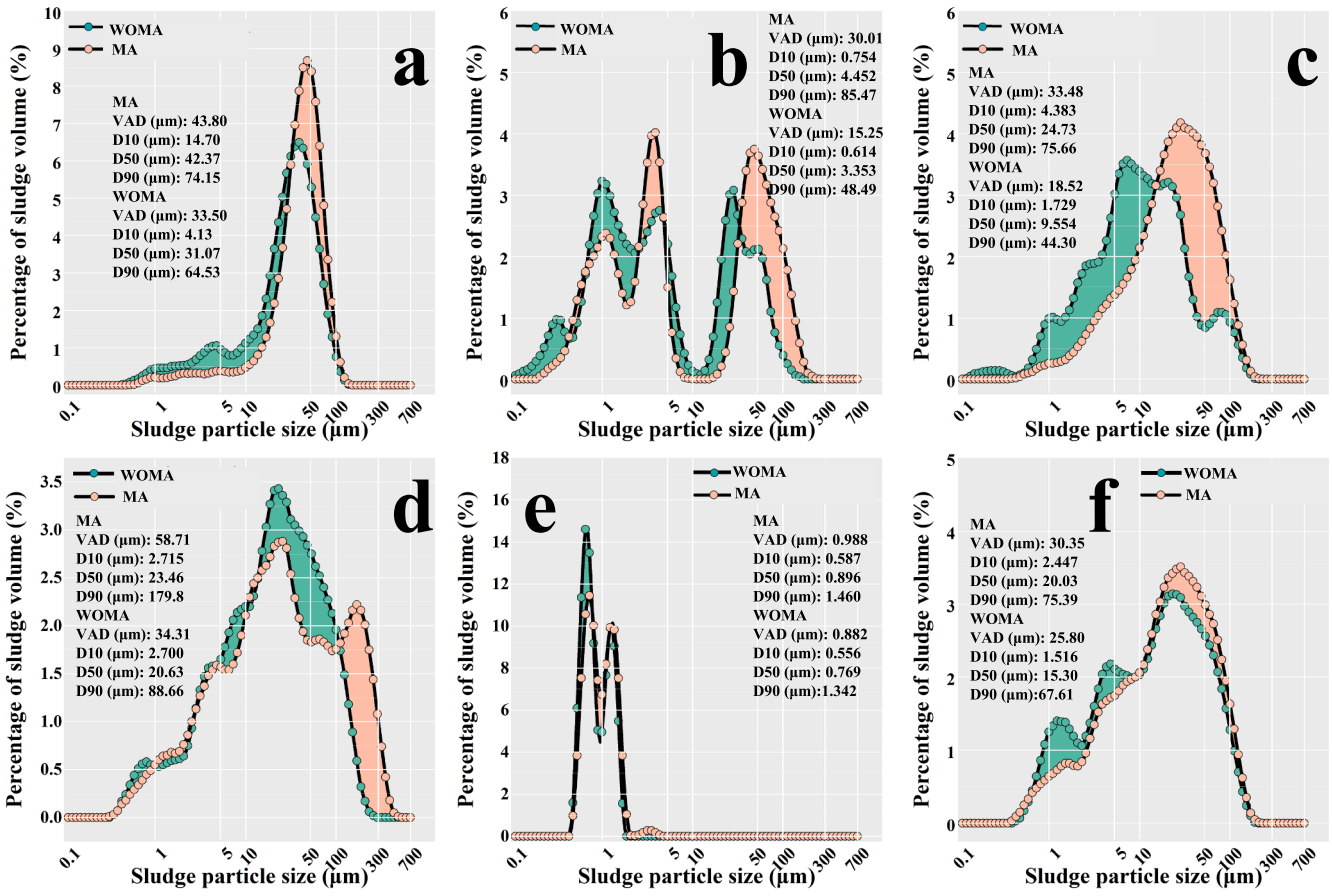


Fig. 4. Particle size distribution of anaerobic sludge (a), difficult-to-settle SS (b) and easy-to-settle SS (c) at the end of AD1 and AD2 for starch-feeding group; and of anaerobic sludge (d), difficult-to-settle SS (e) and easy-to-settle SS (f) at the end of AD4 and AD5 for peptone-feeding group (MA- Microaeration, WOMA- Without microaeration, VAD- Volume average diameter).

Propionibacterium (Xu et al., 2020). Therefore, microaeration could not only enhance the dissolution/hydrolysis of organics, but also greatly reduce the accumulation of propionate and promoted the rapid consumption and utilization of VFAs (Sawatdeenarunat et al., 2017).

3.2. Effect of microaeration on anaerobic sludge filterability

3.2.1. Critical flux of microfiltration membrane

Fig. 3 presents the critical flux measurement results of anaerobic sludge at the end of AD tests. From Fig. 3a and b showing TMP increase with time during critical flux measurement of starch-feeding group, TMP increase rate with time was linearly fitted as 0.0886, 0.124, 0.5301 kPa/min and 0.0349, 0.0819, 0.8368 kPa/min under the constant flux of 1.47, 2.47, 3.44 L/m^2/h (LMH) for no microaeration and microaeration, respectively, indicating a significant improvement of critical flux and anaerobic sludge filterability. From Fig. 3c and d showing TMP increase with time during critical flux measurement of peptone −feeding group, TMP increase rate with time was linearly fitted as 0.0085, 0.1929, 0.2679 kPa/min and 0.0041, 0.0709, 0.3543 kPa/min under the constant flux of 1.76, 2.71, 3.53 LMH for no microaeration and microaeration, respectively. It should be pointed out that some correlation coefficients of linear fitting between TMP and filtration time were not so good (e.g., $R^2 < 0.95$) from Fig. 3. This was mainly due to the complex nature of membrane fouling occurred during a short-term constant-flux filtration of anaerobic sludge (e.g., pore blocking may happen and thus result in that cake/gel layer might not be the dominant fouling mechanism).

For the field of membrane bioreactor with complex sludge mixture (consisting of sludge flocs, colloids, and soluble substances in a large size range of 10^{-3} to 10^{3} μm) to be filtrated, the critical flux is generally regarded as the one, below which no or little TMP increase during a short-term (e.g., 0.5–2 h) constant-flux filtration test (Le Clech et al. 2003a). However, the TMP increasing rate under critical flux has not an accepted standard so far. That's why there are different TMP increasing rate employed to determine critical flux in the literature (De Vela, 2021). Taking TMP increase rate of 0.1 kPa/min as critical flux operation in this study seemed somewhat arbitrary. But it did not affect the data interpretation because it was used to compare the critical flux of anaerobic sludge with and without microaeration while keeping biogas scouring (i.e., representative of crossflow velocity) and membrane the same. Thus, the critical flux zone was determined as 1.76–2.71 LMH and 2.71–3.53 LMH for no microaeration and microaeration, respectively. As for the higher TMP increase rate under the highest flux tested (~3.4–3.5 LMH, already above critical flux) for anaerobic sludge with than without microaeration, it was reasonable because the TMP increase rate during super-critical flux filtration reflected the cake layer resistance increase rate, which was affected by multiple factors of cake layer including thickness, porosity and compressibility (Field & Pearce, 2011). Compared with anaerobic sludge without microaeration, anaerobic sludge with microaeration could have less amount of sludge flocs deposited onto membrane surface but might form cake layer with higher specific resistance, thus leading to a higher TMP increase rate. As a summary, microaeration showed consistent improvements on anaerobic sludge filterability for both starch and peptone feeding groups, which could be attributed to the effects of microaeration on anaerobic sludge characteristics below.

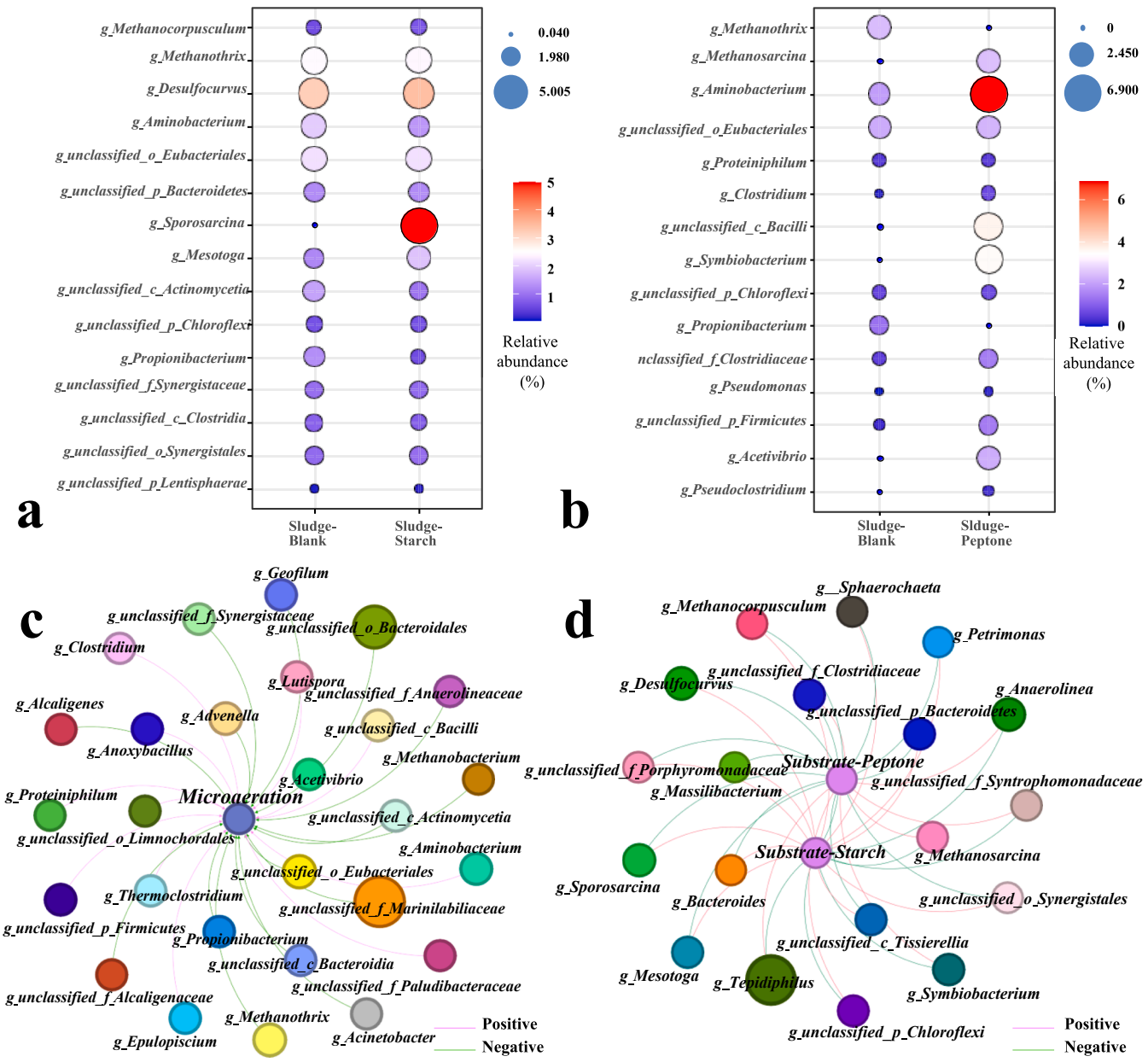


Fig. 5. Species abundance comparison at genus level between Sludge-Starch and Sludge-Blank (a), between Sludge-Peptone and Sludge-Blank (b); network analysis of microaeration and genus (c), network analysis of substrate and genus (d).

3.2.2. Particle size distribution and settleability of anaerobic sludge

From sludge components analysis (see supplementary materials), for starch-feeding group, the easy-to-settle SS accounted for average 50.5 %, 80.7 % and 62.8 % of total SS in AD1, AD2, and AD3, respectively, indicating a significant improvement of settling characteristics by microaeration. For peptone-feeding group, the easy-to-settle SS accounted for average 59.7 %, 63.1 % and 61.7 % of total SS in AD4, AD5, and AD6, respectively, indicating a small improvement of settling characteristics by microaeration. In addition, the higher proportion of easy-to-settle SS in AD3 and AD6 than in AD1 and AD4, respectively, could be derived from attenuation effects of microaeration in AD2 and AD5. Fig. 4 shows the particle size distribution of anaerobic sludge at the end of AD tests. From Fig. 4a and d, particle size distribution of anaerobic sludge showed a clear shift to larger size with average diameter increase from 33.5 to 43.8 μm and 34.3 to 58.7 μm after microaeration for starch-feeding group and peptone-feeding group, respectively, which could result in the proportion increase of easy-to-settle SS (see supplementary materials). According to the critical flux theory and the inertial lift model in crossflow filtration (Field & Pearce, 2011; Huang et al., 2008; Wei et al., 2010), larger particles have higher lateral inertial lift velocity away from membrane surface and thus require higher permeation velocity (i.e., membrane flux) to make them deposited on membrane surface under the same crossflow conditions, resulting in higher critical flux for cake layer fouling formation. This could be the main mechanism behind the higher critical flux and thus better filterability of anaerobic sludge with microaeration in Fig. 3. From Fig. 4b and c, both the easy-to-settle SS and the difficult-to-settle SS from starch-feeding group showed a significant shift to larger size with average diameter increase from 18.5 to 33.5 μm and from 15.3 to 30.0 μm after microaeration, respectively. From Fig. 4e and f, the easy-to-settle SS from peptone-feeding group showed significant shift to larger size with average diameter increase from 25.8 to 30.4 μm while the difficult-to-settle SS showed only marginal size shift with average diameter increase from 0.882 to 0.988 μm after microaeration. Thus, the tendency

to form larger particles induced by microaeration was consistent among the anaerobic sludge and its components (i.e., easy/difficult-to-settle SS).

Previous studies on microaeration for AD mostly focused on biological performance. Only a few studies (Ruan et al., 2019; Zhou et al., 2021) investigated the effects of microaeration on anaerobic sludge dewaterability and found that microaeration could increase particle size and enhance dewaterability. However, there is still no available information on the effects of microaeration on anaerobic sludge filterability for AnMBR application so far in the literature. Although the particle size increase of anaerobic sludge induced by microaeration could provide solid supports for critical flux increase in this study, the deep mechanisms should be further investigated via clarifying how microaeration would affect EPS secretion and the following anaerobic aggregation in the following studies.

3.3. Effect of microaeration and substrate on microbial community and metabolism

3.3.1. Genes abundance encoding enzymes for hydrolysis

The total count of the sequencing data sets for Sludge-Blank, Sludge-Starch and Sludge-Peptone (see deposited raw data referring to accession number of SRR28311681, SRR28311682 and SRR28311683 in NCBI, respectively) was 27023382, 29,678,096 and 28393142, respectively, showing the comparable levels among three samples in this study. The relative abundance (i.e., the count ratio of the specific gene to all genes) was employed for better comparison among the three samples. The abundance of genes encoding enzymes involved in hydrolysis of starch, glucose, and protein revealed the potential effects of microaeration on hydrolysis (see supplementary materials). There was a 6.9 % and 6.7 % higher abundance of genes encoding enzymes involved in hydrolysis of starch and glucose in Sludge-Starch than in Sludge-Blank, respectively. Considering Sludge-Blank had been acclimated to soluble starch (i.e., dissolving the same insoluble starch used in this study in hot water of 70 °C) in AnMBR operation prior to this study, the higher abundance of genes encoding enzymes involved in hydrolysis of starch and glucose in Sludge-Starch could be induced by microaeration, showing the consistency with previous analyses on polysaccharide in Fig. 1a and demonstrating the enhanced hydrolysis by microaeration. There was a 23.3 % higher abundance of genes encoding ATP-dependent protease involved in hydrolysis of protein in Sludge-Peptone than in Sludge-Blank. Considering Sludge-Blank had not been acclimated to peptone, the higher abundance of genes encoding enzymes involved in hydrolysis of protein in Sludge-Peptone could be derived from both acclimation and microaeration. These results demonstrated that microaeration could enhance hydrolysis of polysaccharide and protein via promoting the secretion of hydrolase (Fu et al., 2022; Nguyen & Khanal, 2018).

3.3.2. Species abundance at genus level

Fig. 5a and b show the species abundance (at genus level) comparison between Sludge-Starch and Sludge-Blank, and between Sludge-Peptone and Sludge-Blank. First of all, as a fast growing and producing propionic acid species under exclusively anaerobic conditions (Collograi et al., 2022), *Propionibacterium* showed an abundance of 1.3 % in Sludge-Blank while dramatically decreased to an abundance of 0.56 % and 0.00024 % in Sludge-Starch and Sludge-Peptone, respectively. This could be derived from the inhibition of microaeration on *Propionibacterium* in Sludge-Starch and Sludge-Peptone, thus leading to lower abundance of *Propionibacterium* and propionic acid production, which was consistent with the previous results of VFAs in Fig. 2. These results demonstrated that microaeration could enhance AD performance via reducing *Propionibacterium* abundance and thus propionic acid production.

From Fig. 5a, *Mesotoga* involved in acetate oxidation (Meng et al., 2022), was enriched from the abundance of 1.21 % in Sludge-Blank (previously acclimated to soluble starch) to the abundance of 1.87 % in Sludge-Starch under microaeration. This could be ascribed into microaeration and result in the low acetic acid at the end of AD2 (shown in Fig. 4a). *Desulfocurvus*, which could facilitate the dissolution/hydrolysis of organics, degrade organics as iron reducing bacteria, and establish direct interspecies electron transfer (DIET) with some methanogens to improve methanogenic efficiency (Xu et al., 2020), was enriched from the abundance of 3.21 % in Sludge-Blank to the abundance of 3.43 % in Sludge-Starch, indicating the enhanced AD performance by microaeration. *Sporosarcina* mainly involved in fermentative acidification including acidogenesis and acetogenesis (Mahdy et al., 2019; Thakur et al., 2024), was enriched from the abundance of 0.04 % in Sludge-Blank to the abundance of 5.01 % in Sludge-Starch, indicating the significantly enhanced fermentative acidification by microaeration. For methanogenic species, *Methanothrix* and *Methanocorpusculum* widely known to be involved in DIET (Wang et al., 2023; Xu et al., 2020), showed the abundance of 2.45 % and 0.58 % in Sludge-Blank and the abundance of 2.41 % and 0.68 % in Sludge-Starch, respectively, demonstrating that the microaeration did not only inhibit methanogens but also enrich methanogens via promoting dissolution/hydrolysis and DIET in this study. Considering Sludge-Blank had been acclimated to soluble starch for a long time prior to this study, it was reasonable that there was a similar microbial community structure between Sludge-Blank and Sludge-Starch.

From Fig. 5b, the species involved in the hydrolysis/acidification of organics including *Pseudomonas*, *Firmicutes*, *Clostridiaceae*, *Chloroflexi*, *Bacilli*, and *Clostridium* (Gamble et al., 2011; Ye et al., 2017), were all enriched from the abundance of 0.07 %, 0.27 %, 0.53 %, 0.62 %, 0.01 %, 0.13 % in Sludge-Blank to the abundance of 0.16 %, 1.47 %, 1.48 %, 0.71 %, 3.69 %, 0.67 % in Sludge-Peptone, respectively. *Proteiniphilum* responsible for protein hydrolysis (Tao et al., 2023), was slightly enriched from the abundance of 0.44 % in Sludge-Blank to the abundance of 0.46 % in Sludge-Peptone. *Aminobacterium*, which could oxidize amino acids (e.g., alanine, glutamate, leucine, isoleucine, valine, aspartate, and methionine) to produce formate for methanogenesis (Baena et al., 2000; Baena et al., 1998; Hamdi et al., 2015), was significantly enriched from the abundance of 1.98 % in Sludge-Blank to the abundance of 6.90 % in Sludge-Peptone. For methanogenic species, *Methanothrix* and *Methanosarcina* showed the abundance of 2.45 % and 0.01 % in Sludge-Blank, and of 0.0012 % and 2.49 % in Sludge-Peptone, respectively, indicating a clear transition from dominant *Methanothrix* in Sludge-Blank to *Methanosarcina* in Sludge-Peptone. *Methanothrix* is known to use only acetic acid to produce methane, while *Methanosarcina* can achieve methane production using multiple substrates of methylamine, methanol, hydrogen and acetic acid (Guo et al., 2015). This transition could be a reason for the higher methane production. Considering the Sludge-Blank had not been acclimated to peptone in this study, the significant difference of microbial community structure between Sludge-Blank and Sludge-Peptone could be derived from the effects of microaeration as well as substrate transfer from starch to peptone.

Fig. 5c and d show the correlation network between environmental factors and species. Consistent with the abovementioned results, there was a positive correlation between microaeration and most facultative bacteria in Fig. 5c, which agreed well with some previous studies (Amin et al., 2021; Fu et al., 2022). However, there was a negative correlation between microaeration and some methanogens, indicating that microaeration was not actually favorable to the growth of methanogens. Therefore, the abundance increase of some methanogens in AD with microaeration in this study could be ascribed into 1) preventing methanogens from oxygen exposure via rapid oxygen consumption by facultative bacteria and formation of large layered particles (Nguyen & Khanal, 2018), and 2) providing sufficient substrates for methanogens via promoting dissolution/hydrolysis and DIET. In addition, microaeration showed a positive effect on Proteiniphilum and Aminobacterium, which could be one of potential reasons for the enhance protein

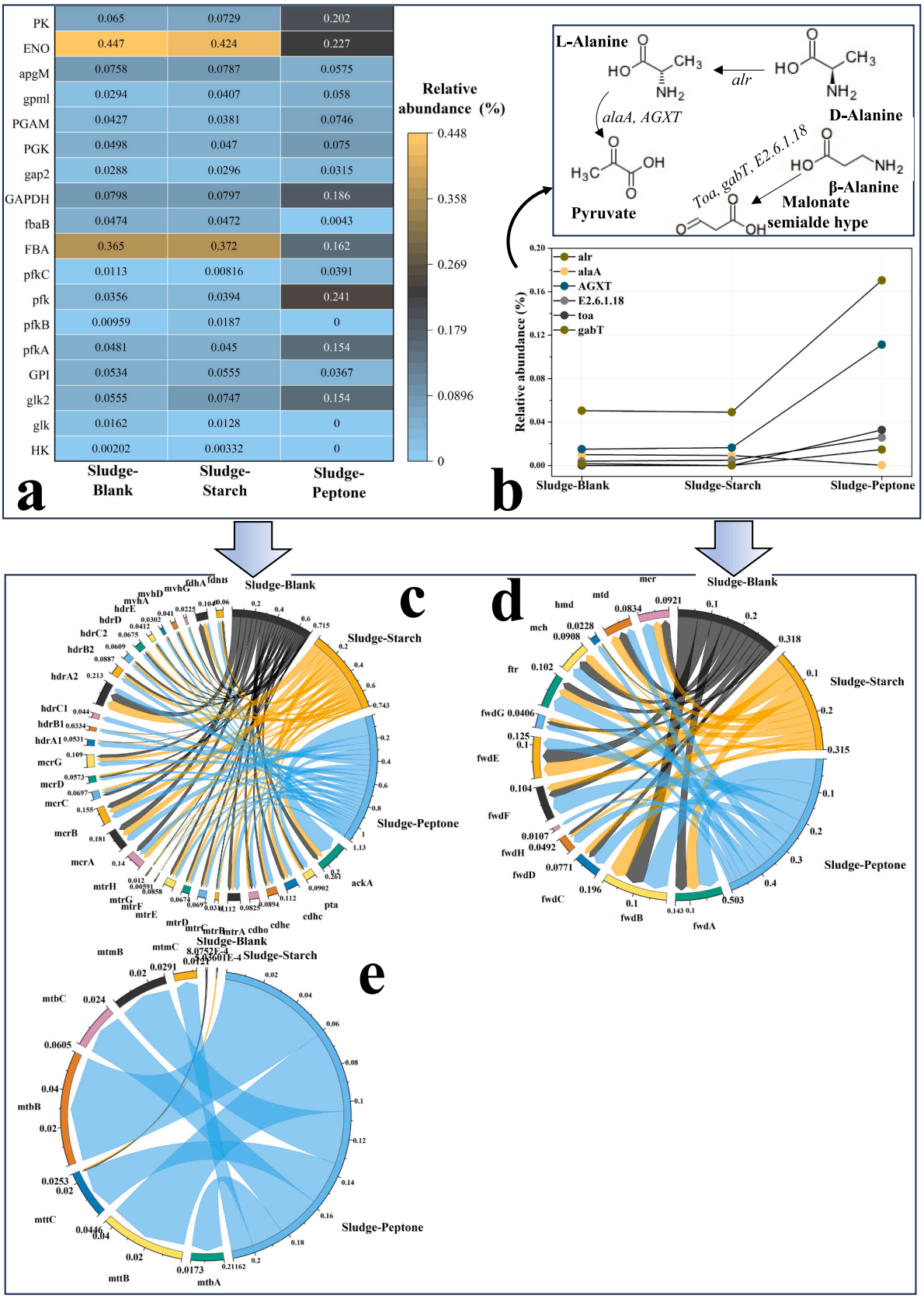


Fig. 6. Abundance of genes involved in glycolysis (a), alanine (e.g., one typical amino acid) metabolism (b), methane metabolism via acetate-type (c), carbon dioxide-type (d), methyl compound-type (e).

hydrolysis and amino acid oxidation. From Fig. 5d, Methanosarcina and Symbiobacterium showed negative correlation in Sludge-Starch and positive correlation in Sludge-Peptone, while Sporosarcina and Desulfocurvus showed positive correlation in Sludge-Starch negative and correlation in Sludge-Peptone, revealing the important effects of substrates selective pressure on specific microbes.

3.3.3. Genes abundance involved in anaerobic metabolism

Enrichment of functional flora usually increases metabolic activity. Fig. 6 presents the abundance of functional genes involved in glycolysis (Fig. 6a), amino acid metabolism (Fig. 6b), methane metabolism via acetate-type (Fig. 6c), carbon dioxide-type (Fig. 6d) and methyl compound-type (Fig. 6e). From Fig. 6a**,** Sludge-Starch showed the 12.2 %, 2 %, 3.9 % higher abundance of *PK, FBA*, *GPI*, and the 5.43 %, 5.79 %, 6.72 % lower abundance of *ENO*, *PGK*, *pfkA*, respectively, than Sludge-Blank, indicating the minor effects of microaeration on glycolysis. Compared with Sludge-Blank, Sludge-Peptone showed the 211 %, 74.6 %, 132.6 %, 576.3 %, 219.5 %, 177.7 % higher abundance of *PK, PGAM, GAPDH, pfk, pfkA glk2*, and the 96.9 %, 32 %, 125 % lower abundance of *ENO, apgM, FBA*, respectively, and absence of some genes (*pfkB, glk, HK*), indicating the major effects of substrate transfer from starch to peptone on glycolysis. From Fig. 6b, Sludge-Starch showed the 1.87 % lower abundance of the functional genes involved in alanine (one of typical amino acids) metabolism than Sludge-Blank, which might be caused by the lack of amino acid source from the substrate. Sludge-Peptone showed the 338.5 % higher abundance of the functional genes involved in alanine metabolism than Sludge-Blank, which could be derived from substrate transfer from starch to peptone as well as microaeration, considering the positive correlation between microaeration and *Aminobacterium* responsible for amino acids metabolism in Fig. 5c.

From Fig. 6c, d, and e**,** Sludge-Blank, Sludge-Starch, Sludge-Peptone showed the abundance of 0.715 %, 0.743 %, 1.13 % for genes involved in the acetate-type methanogenic pathway, the abundance of 0.318 %, 0.315 %, 0.503 % for genes involved in carbon dioxide-type methanogenic pathway, and the abundance of 0.000808 %, 0.000504 %, 0.212 % for genes involved in methyl compound-type methanogenic pathway, respectively. The very low abundance of genes involved in methyl compound-type methanogenic pathway in Sludge-Blank and Sludge-Starch could be due to the lack of methyl compound (e.g., methylamine) and its precursor (e.g., amino acid) caused by the lack of protein in the substrate (Guo et al., 2015), which was consistent with the low abundance of *Methanosarcina* in Sludge-Blank and Sludge-Starch in Fig. 5. The similar abundance of genes involved in the acetate-type and carbon dioxide-type methanogenic pathways between Sludge-Starch and Sludge-Blank indicated no obvious negative effects of microaeration on methane metabolism. Sludge-Peptone showed the 261.1 % higher abundance of genes involved in methyl compound-type methanogenic pathway than Sludge-Blank, indicating the strong effects of substrate transfer from starch to peptone on methane metabolism, which was consistent with the 248 % higher abundance of *Methanosarcina* in Sludge-Peptone than Sludge-Blank in Fig. 5. In addition, for all three samples, there was a descending order for the abundance of genes involved in the acetate-type, carbon dioxide-type, and methyl compound-type methanogenic pathways, indicating the dominant acetate-type and carbon dioxide-type methanogenic pathways in this study.

4. Conclusions

Bubble-free membrane microaeration was employed to enhance dissolution/ hydrolysis of insoluble/macromolecular organics and sludge filterability for AnMBR application. The batch AD tests feeding insoluble starch and soluble peptone with and without microaeration showed that microaeration increased methane production via promoting dissolution/hydrolysis/acidification and critical flux of microfiltration membrane via driving large sludge flocs formation. The metagenomic analyses showed that microaeration enriched functional genes and bacteria for hydrolysis and acidification but showed little effect on functional genes for anaerobic metabolisms. The optimal ORP for microaeration-assisted AnMBR and evaluation of the promising microaeration method via bubble-free membrane or nanobubble are worthy of investigation.

CRediT authorship contribution statement

Chun-Hai Wei: Writing – original draft, Supervision, Investigation, Funding acquisition, Conceptualization. **Zheng-Wei Wang:** Writing – original draft, Visualization, Investigation, Funding acquisition. **Jian-Hua Dai:** Writing – original draft, Investigation. **Kang Xiao:** Writing – review & editing, Supervision, Funding acquisition. **Hua-Rong Yu:** Writing – review & editing, Funding acquisition. **Fang-Shu Qu:** Writing – review & editing, Funding acquisition. **Hong-Wei Rong:** Resources. **Jun-Guo He:** Resources. **Huu Hao Ngo:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This study was jointly funded by the Guangdong Basic and Applied Basic Research Foundation (2024A1515011347), National Natural Science Foundation of China (52070045, 52170072, 52370059), Foreign Expert Program (G2022031007L) and Graduate Student Innovation Ability Training Funding Program of Guangzhou University (2022GDJC-M40). The authors also appreciate Shiyanjia Lab (www.shiyanjia.com) and Majorbio Bio-Pharm Technology (www.majorbio.com) for analytical assistance.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2024.130787.

References

Amin, F.R., Khalid, H., Li, W., Chen, C., Liu, G., 2021. Enhanced methane production and energy potential from rice straw by employing microaerobic pretreatment via anaerobic digestion. J. Clean. Prod. 296, 126434.

Aslam, A., Khan, S.J., Shahzad, H.M.A., 2021. Anaerobic membrane bioreactors (AnMBRs) for municipal wastewater treatment- potential benefits, constraints, and future perspectives: An updated review. Sci. Total Environ. 802, 149612.

Baena, S., Fardeau, M.L., Labat, M., Ollivier, B., Thomas, P., Garcia, J.L., Patel, B.K.C., 1998. Aminobacterium colombiensegen. nov. sp. nov., an Amino Acid-degrading Anaerobe Isolated from Anaerobic Sludge. Anaerobe 4 (5), 241–250.

Baena, S., Fardeau, M.L., Labat, M., Ollivier, B., Garcia, J.L., Patel, B.K., 2000. Aminobacterium mobile sp. nov., a new anaerobic amino-acid-degrading bacterium. Int. J. Syst. Evol. Microbiol. 50 (1), 259–264.

Cao, Q., Zhang, W., Lian, T., Wang, S., Yin, F., Zhou, T., Wei, X., Dong, H., 2022. Revealing mechanism of micro-aeration for enhancing volatile fatty acids production from swine manure. Bioresour. Technol. 365, 128140.

Chen, Q., Wu, W., Qi, D., Ding, Y., Zhao, Z., 2019. Review on microaeration-based anaerobic digestion: State of the art, challenges, and prospectives. Sci. Total Environ. 710, 136388.

Cheng, Z., Wei, Y., Zhang, Q., Zhang, J., Lu, T., Pei, Y., 2018. Enhancement of surfactant biodegradation with an anaerobic membrane bioreactor by introducing microaeration. Chemosphere 208, 343–351.

Collograi, K.C., da Costa, A.C., Ienczak, J.L., 2022. Fermentation strategies to improve propionic acid production with propionibacterium ssp.: a review. Crit. Rev. Biotechnol. 42 (8), 1157–1179.
De Vela, R.J., 2021. A review of the factors affecting the performance of anaerobic membrane bioreactor and strategies to control membrane fouling. Reviews in Environmental Science and Bio/technology 20 (3), 607–644.
Federation, W.E., Association, A., 2005. Standard methods for the examination of water and wastewater. American Public Health Association (APHA), Washington, DC, USA, p. 21.
Field, R.W., Pearce, G.K., 2011. Critical, sustainable and threshold fluxes for membrane filtration with water industry applications. Adv. Colloid Interface Sci. 164 (1), 38–44.
Fox, R.A., Stuckey, D.C., 2015. The effect of sparging rate on transmembrane pressure and critical flux in an AnMBR. J. Environ. Manage. 151, 280–285.
Fu, S.-F., Chen, K.-Q., Sun, W.-X., Zhu, R., Zheng, Y., Zou, H., 2018. Improved methane production of corn straw by the stimulation of calcium peroxide. Energ. Conver. Manage. 164, 36–41.
Fu, S., Lian, S., Angelidaki, I., Guo, R., 2022. Micro-aeration: an attractive strategy to facilitate anaerobic digestion. Trends Biotechnol. 41 (5), 714–726.
Gamble, M., Künze, G., Dodson, E.J., Wilson, K.S., Jones, D.D., 2011. Regulation of an intracellular subtilisin protease activity by a short propeptide sequence through an original combined dual mechanism. PNAS 108 (9), 3536–3541.
Guo, J., Peng, Y., Ni, B.-J., Han, X., Fan, L., Yuan, Z., 2015. Dissecting microbial community structure and methane-producing pathways of a full-scale anaerobic reactor digesting activated sludge from wastewater treatment by metagenomic sequencing. Microb. Cell Fact. 14, 33.
Hamdi, O., Ben Hania, W., Postec, A., Bouallagui, H., Hamdi, M., Bonin, P., Ollivier, B., Fardeau, M.-L., 2015. Aminobacterium thunnarium sp. nov., a mesophilic, amino acid-degrading bacterium isolated from an anaerobic sludge digester, pertaining to the phylum Synergistetes. Int. J. Syst. Evol. Microbiol. 65 (Pt_2), 609–614.
Hu, Y., Wang, X.C., Ngo, H.H., Sun, Q., Yang, Y., 2017. Anaerobic dynamic membrane bioreactor (AnDMBR) for wastewater treatment: A review. Bioresour. Technol. 247, 1107–1118.
Huang, X., Wei, C.-H., Yu, K.-C., 2008. Mechanism of membrane fouling control by suspended carriers in a submerged membrane bioreactor. J. Membr. Sci. 309 (1–2), 7–16.
Iftikhar, Z., Sana, Y., Muhammad, A., Aqsa, Y., Anwar Zeb, K., Junaid Ahmad, T., Quanbao, Z., Arshad Mehmood, A., Raza, A., Tariq Mahmood, K., Asim, Y., Muhammad, B., 2021. In-situ microaeration of anaerobic digester treating buffalo manure for enhanced biogas yield. Renew. Energy 181, 843–850.
Le Clech, P., Jefferson, B., Chang, I.S., Judd, S.J., 2003. Critical flux determination by the flux-step method in a submerged membrane bioreactor. J. Membr. Sci. 227 (1–2), 81–93.
Liu, X.-W., Sheng, G.-P., Yu, H.-Q., 2009. Physicochemical characteristics of microbial granules. Biotechnol. Adv. 27 (6), 1061–1070.
Mahdy, A., Wandera, S.M., Bi, S., Song, Y., Qiao, W., Dong, R., 2019. Response of the microbial community to the methanogenic performance of biologically hydrolyzed sewage sludge with variable hydraulic retention times. Bioresour. Technol. 288, 121581.
Meng, X., Cao, Q., Sun, Y., Huang, S., Liu, X., Li, D., 2022. 16S rRNA genes- and metagenome-based confirmation of syntrophic butyrate-oxidizing methanogenesis enriched in high butyrate loading. Bioresour. Technol. 345, 126483.
Nguyen, D., Khanal, S.K., 2018. A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. Biotechnol. Adv. 36 (7), 1971–1983.
Ruan, D., Zhou, Z., Pang, H., Yao, J., Chen, G., Qiu, Z., 2019. Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis. Bioresour. Technol. 289, 121643.
Sawatdeenarunat, C., Sung, S., Khanal, S.K., 2017. Enhanced volatile fatty acids production during anaerobic digestion of lignocellulosic biomass via micro-oxygenation. Bioresour. Technol. 237, 139–145.
Song, X., Luo, W., Hai, F.I., Price, W.E., Guo, W., Ngo, H.H., Nghiem, L.D., 2018. Resource recovery from wastewater by anaerobic membrane bioreactors: opportunities and challenges. Bioresour. Technol. 270, 669–677.
Stewart, P.S., Franklin, M.J., 2008. Physiological heterogeneity in biofilms. Nat. Rev. Microbiol. 6 (3), 199–210.
Tao, L., Zhi-Jing, X., Ning-Yi, Z., 2023. Aerobic degradation of the antidiabetic drug metformin by aminobacter sp. Strain NyZ550. Environ. Sci. Tech. 57 (3), 1510–1519.
Tartakovsky, B., Mehta, P., Bourque, J.S., Guiot, S.R., 2011. Electrolysis-enhanced anaerobic digestion of wastewater. Bioresour. Technol. 102 (10), 5685–5691.
Thakur, N., Jalalah, M., Alsareii, S.A., Harraz, F.A., Almadiy, A.A., Su, S., Salama, E.-S., Li, X., 2024. Anaerobic digestion of fat, oil, and grease (FOG) under combined additives: Enhanced digestibility, biogas production, and microbiome. Renew. Sustain. Energy Rev. 191, 114155.
Wang, Z.-W., Chun-Hai, W., Hua-Rong, Y., Fang-Shu, Q., Hong-Wei, R., Jun-Guo, H., Guang-Li, L., Xia, H., Hao, N.H., 2023. Preparation and mechanism of carbon felt supported iron trioxide and zero-valent iron for enhancing anaerobic digestion performance. Chem. Eng. J. 468, 143565.
Wei, C.-H., Huang, X., Ben Aim, R., Yamamoto, K., Amy, G., 2010. Critical flux and chemical cleaning-in-place during the long-term operation of a pilot-scale submerged membrane bioreactor for municipal wastewater treatment. Water Res. 45 (2), 863–871.
Wu, Z., Nguyen, D., Lam, T.Y.C., Zhuang, H., Shrestha, S., Raskin, L., Khanal, S.K., Lee, P.-H., 2020. Synergistic association between cytochrome bd-encoded Proteiniphilum and reactive oxygen species (ROS)-scavenging methanogens in microaerobic-anaerobic digestion of lignocellulosic biomass. Water Res. 190, 116721.
Wu, Z., Nguyen, D., Shrestha, S., Raskin, L., Khanal, S.K., Lee, P.-H., 2023. Evaluation of nanaerobic digestion as a mechanism to explain surplus methane production in animal rumina and engineered digesters. Environ. Sci. Tech. 57 (33), 12302–12314.
Xu, Y., Wang, M., Yu, Q., Zhang, Y., 2020. Enhancing methanogenesis from anaerobic digestion of propionate with addition of Fe oxides supported on conductive carbon cloth. Bioresour. Technol. 302, 122796.
Ye, J., Hu, A., Ren, G., Zhou, T., Zhang, G., Zhou, S., 2017. Red mud enhances methanogenesis with the simultaneous improvement of hydrolysis-acidification and electrical conductivity. Bioresour. Technol. 247, 131–137.
Ying, L., Liguo, S., Zhengyi, H., Jiahao, L., Yanchao, X., Renjie, L., Meijia, Z., Huachang, H., Hongjun, L., 2021. A novel in-situ micro-aeration functional membrane with excellent decoloration efficiency and antifouling performance. J. Membr. Sci. 641, 119925.
Yu, D., Liu, J., Sui, Q., Wei, Y., 2016. Biogas-pH automation control strategy for optimizing organic loading rate of anaerobic membrane bioreactor treating high COD wastewater. Bioresour. Technol. 203, 62–70.
Zhang, N., Liu, W., Peng, Y., Song, X., 2021a. Anaerobic membrane bioreactors for livestock wastewater treatment and resource recovery: opportunities and challenges. Current Pollution Reports 7 (3), 277–285.
Zhang, L., Loh, K.-C., Lim, J.W., Zhang, J., 2019. Bioinformatics analysis of metagenomics data of biogas-producing microbial communities in anaerobic digesters: A review. Renew. Sustain. Energy Rev. 100, 110–126.
Zhang, D.-M., Teng, Q., Zhang, D., Jilani, G., Ken, W.-M., Yang, Z.-P., Alam, T., Ikram, M., Iqbal, Z., 2020. Performance and microbial community dynamics in anaerobic continuously stirred tank reactor and sequencing batch reactor (CSTR-SBR) coupled with magnesium-ammonium-phosphate (MAP)-precipitation for treating swine wastewater. Bioresour. Technol. 320, 124336.
Zhang, Z., Yin, Y., Hongbo, X., Yuexi, Z., 2021b. Review of micro-aeration hydrolysis acidification for the pretreatment of toxic and refractory organic wastewater. J. Clean. Prod. 317, 128343.
Zhang, Z., Xi, H., Yu, Y., Wu, C., Yang, Y., Guo, Z., Zhou, Y., 2022. Coupling of membrane-based bubbleless micro-aeration for 2,4-dinitrophenol degradation in a hydrolysis acidification reactor. Water Res. 212, 118119.
Zhou, Z., Ming, Q., An, Y., Ruan, D., Chen, G., Wei, H., Wang, M., Wu, Z., 2021. Performance and microbial community analysis of anaerobic sludge digestion enhanced by in-situ microaeration. J. Water Process Eng. 42, 102171.